2025 NATIONAL RESEARCH CONFERENCE FOR THE PREVENTION OF FIREARM-RELATED HARMS

Beyond Urban–Rural Binaries: Remote Sensing Indicators of Built Environment Risk for Firearm Injury in Cook County, Illinois

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Background

- Firearm-related death is a pervasive public health crisis across the U.S., affecting communities across the urban–rural spectrum
- While total rates of firearm mortality are comparable across the urban–rural continuum, the mechanism of death varies:
  - Urban areas → higher rates of firearm assault (homicide)
  - Rural areas → higher rates of firearm self-harm (suicide)
- Temporal patterns
  - From 2000 to 2020, rural counties saw growing firearm mortality rates, overtaking urban areas in several regions
- Despite these shifting spatiotemporal patterns, research continues to rely on static, county-level classifications of urbanicity and rurality, which obscure within-area variation in deprivation, density, and environmental exposure

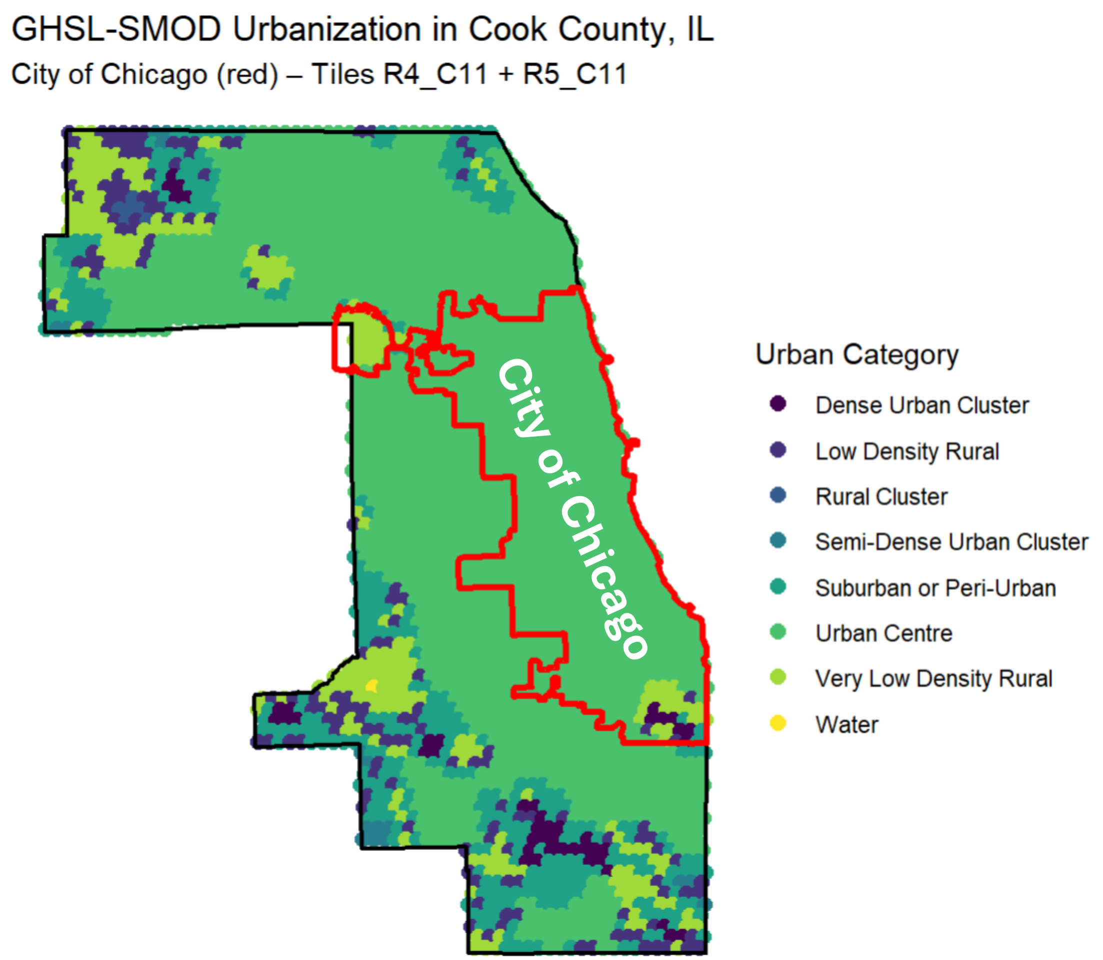
Bridging the Gap: Beyond Urban/Rural Binaries

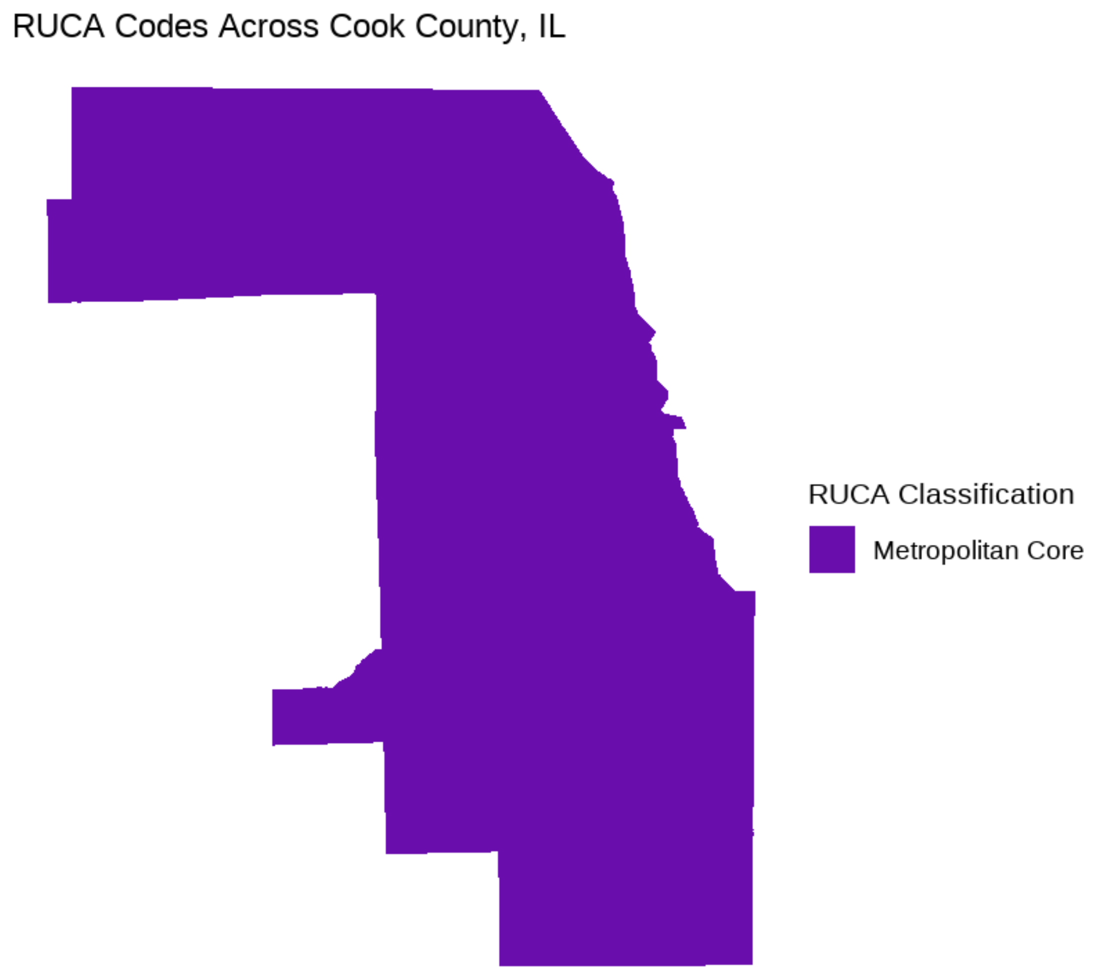
- Traditional urban–rural classifications (e.g., USDA, NCHS, RUCA):
  - Simplify complex settlement patterns
  - Obscure local variation within counties and metropolitan regions
  - Rely on administrative boundaries
  - Introduce ecological bias
- This study bridges those limitations by integrating areal-level social deprivation (ADI) with high-resolution, continuous geospatial indicators of the built environment:
  - Built-up Intensity – density of constructed surfaces
  - Nighttime Light (VIIRS)
  - Vegetative Greenness (NDVI)
  - Area Deprivation Index (ADI)

GHS-SMOD vs. Conventional Urban–Rural Codes: A Higher-Resolution Approach to Urban Form

Dimension	GHS-SMOD (This Study)	RUCA Codes (Conventional)
Source Data	Satellite imagery + population grids (Landsat, Sentinel, GHS-POP)	Census commuting flows between tracts
Unit of Geography	Census Block Group (resampled from 1km raster)	Census Tract (U.S. only)
Classification Basis	Built-up intensity + population density (9 types)	Work commuting patterns + population thresholds
Spatial Resolution	High (sub-county scale; detects transitions between dense and low-density zones), reveals w/in county gradients	Moderate (tract-level only); Coarse thresholds may mask inequities
Temporal Updates	Consistent global data every ~5 years (1975–2020)	Updated only after decennial Census (every 10 years)
Focus / Model	Captures how space is built, inhabited, and changing over time	Captures where people work and live

Urbanization Typology in Cook County (GHS v RUCA)



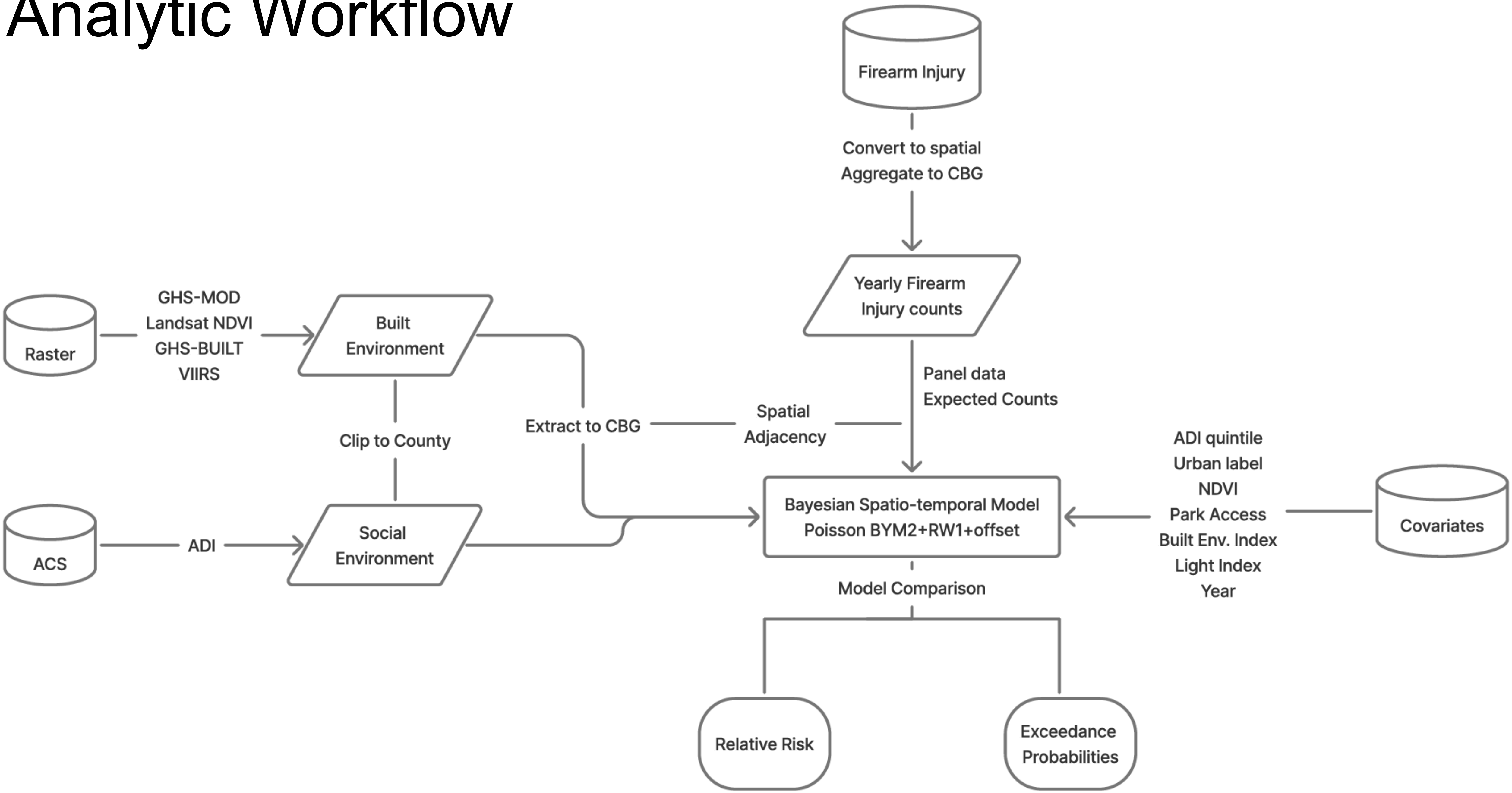


Fine-grained satellite typology (left) reveals intra-county variation invisible in conventional RUCA codes (right).

Other Study Variables

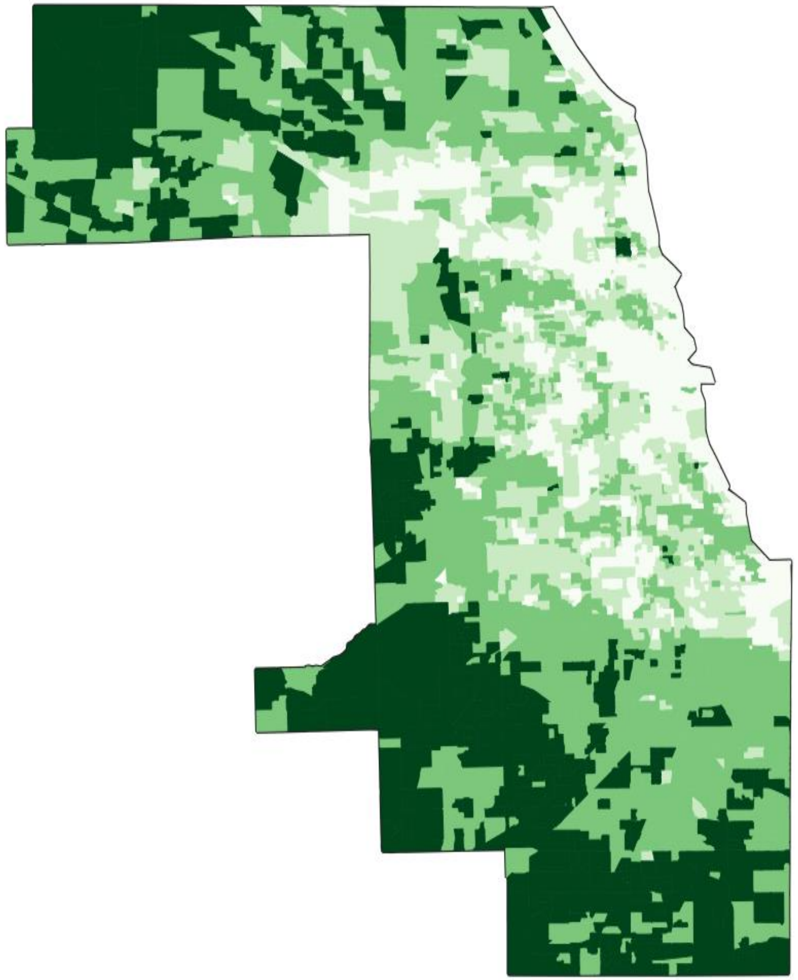
Domain	Data Source	Indicator	What It Adds
Firearm Injury	Chicago Open Data Portal	Incident locations → CBG-level counts	Micro-scale spatial precision
Built Environment	GHS-BUILT (Global Human Settlement)	Built-up intensity	Measures built density and urban form
Human Activity	VIIRS Nighttime Lights	Light intensity	Proxy for nighttime human activity and exposure
Vegetation	Landsat 8–9	NDVI (Greenness)	Captures land cover and ecological context
Socioeconomic	Area Deprivation Index (ADI)	ADI Quintiles	Captures neighborhood-level deprivation and structural inequality
Park Access	Transit Equity Dashboard	Park acreage reachable within 30 minutes by public transit	Measures accessibility to green space and public amenities via transit

Analytic Workflow

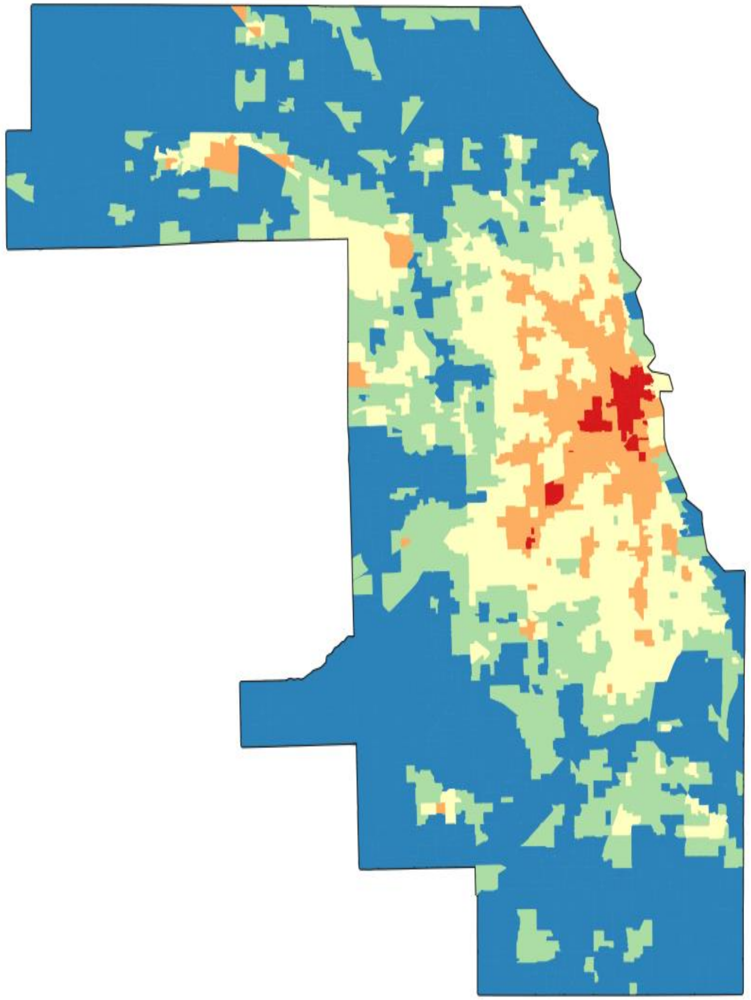


High-resolution environmental rasters (NDVI, VIIRS nighttime lights, and GHS-BUILT) were clipped to Cook County and aggregated to census block groups.

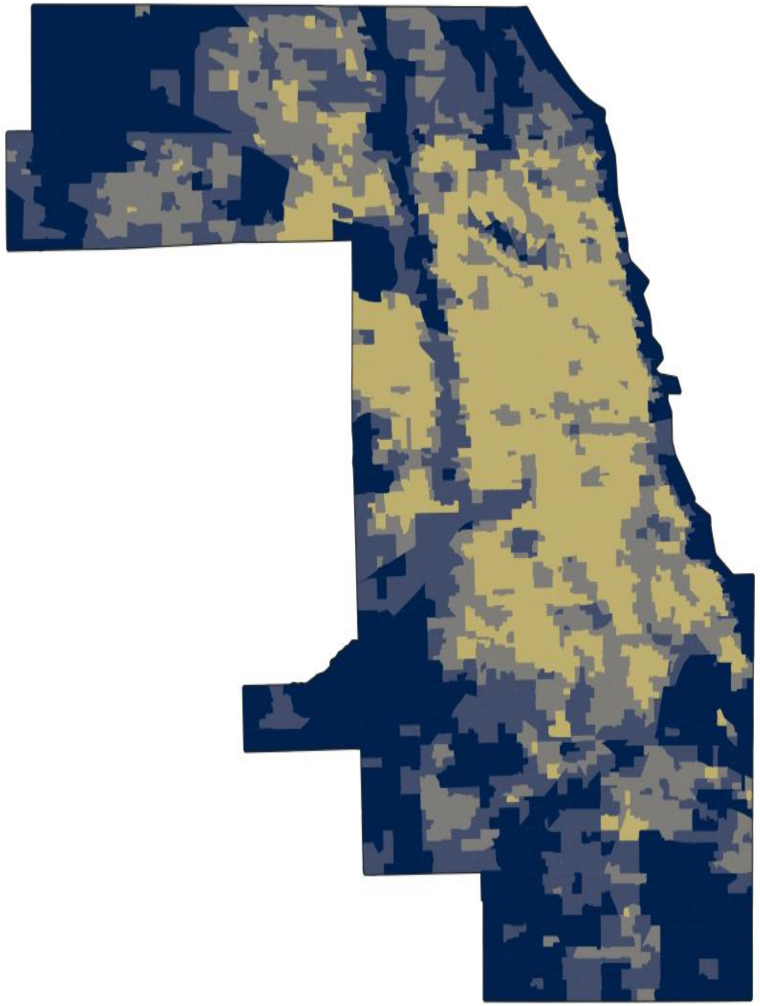
Vegetation (NDVI)

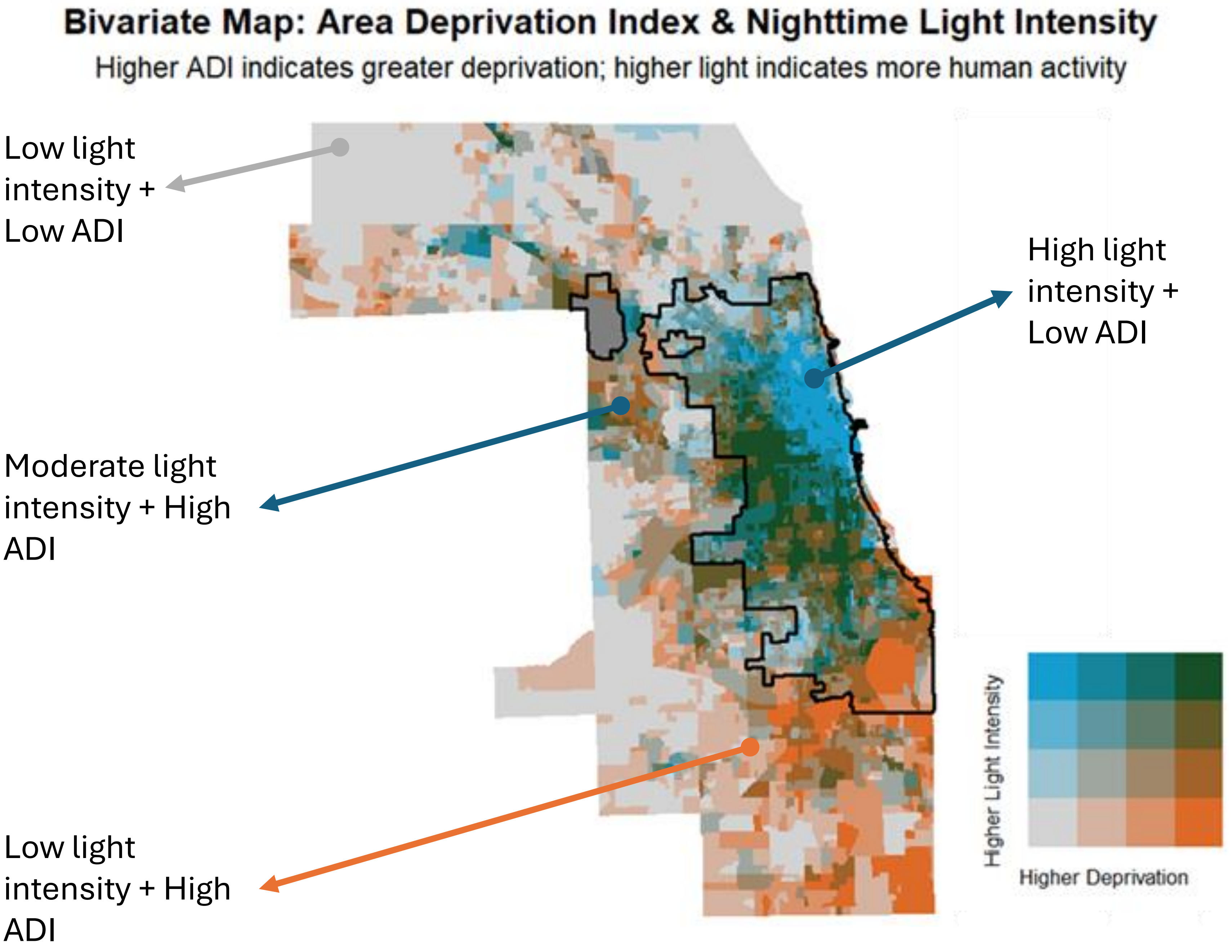


Light Exposure (VIIRS)



Built Environment (GHS-BUILT)



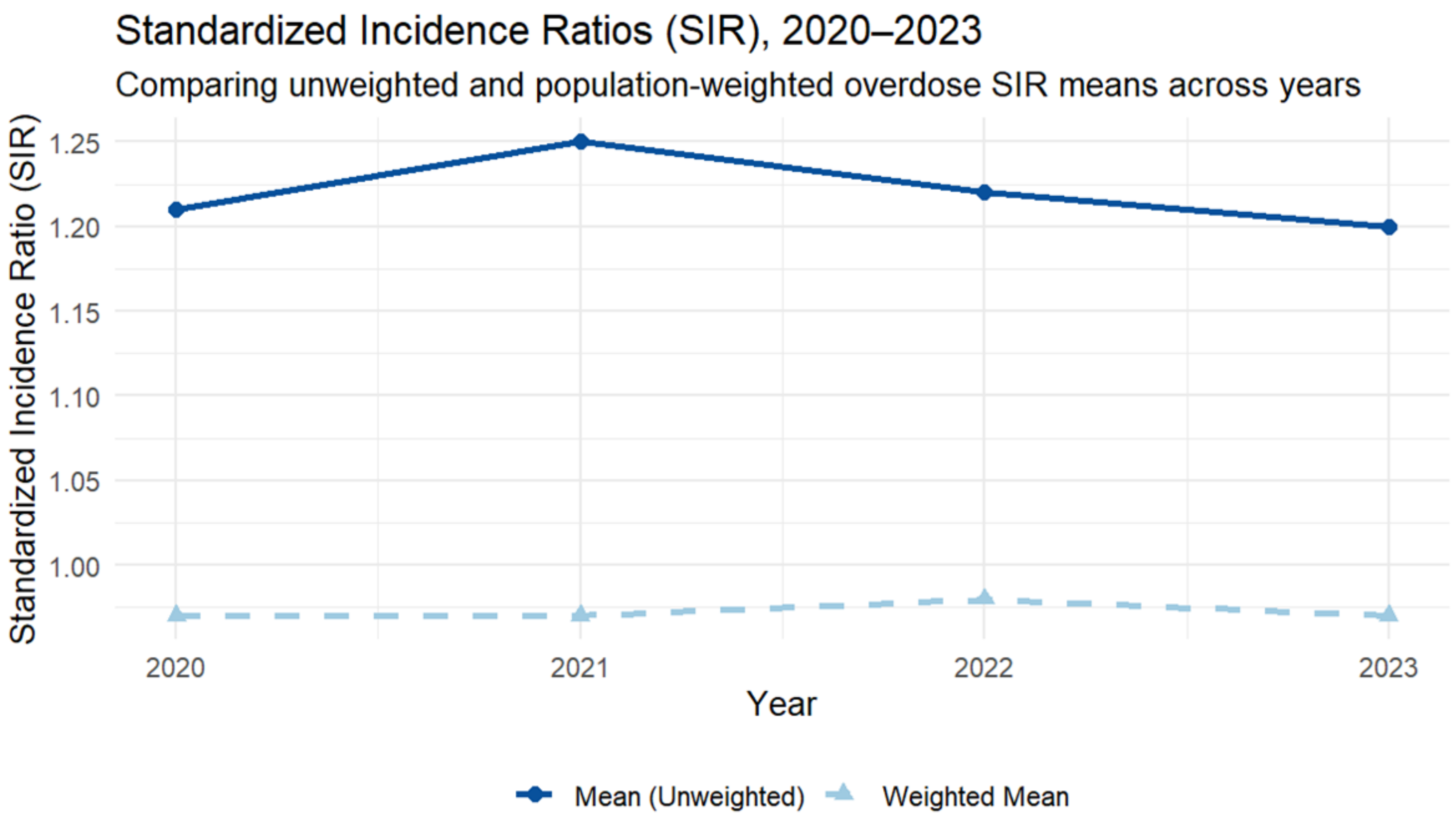


- Combines satellite-derived nighttime light intensity (VIIRS) with neighborhood deprivation (ADI) to visualize overlapping social and environmental inequalities
- Higher light intensity (top of legend) indicates greater human activity or urbanization; higher deprivation (right of legend) reflects greater socioeconomic disadvantage.
- Visual evidence further supporting that binary rural-urban codes miss shared overlapping risks

Neighborhood Contexts of Firearm-Related Harm

	Firearm-Related Homicide (n = 3,208)	Firearm-Related Suicide (n = 661)
Sex		
Male	89.6	89.3
Female	10.4	10.7
Race/Ethnicity		
NH Black	79.5	27.5
NH White	3.3	55.1
Hispanic	16.4	14.7
NH Asian	0.5	1.7
Age	30.4 ± 11.6	47.1 ± 20.5
Context		
ADI	119.98 ± 16.71	100.79 ± 18.12
NDVI	0.13 ± 0.09	0.16 ± 0.1
Built	0.29 ± 0.05	0.27 ± 0.07
Light	0.25 ± 0.08	0.19 ± 0.11
Acres	48.73 ± 120.42	101.98 ± 283.36
Urban Label	Urban Center	Urban Center

Neighborhood-Level Variation in Firearm Injury Risk



Statistic	What It Reflects	Practical Interpretation
Unweighted Mean SIR = 1.20	Average neighborhood-level excess risk	Some tracts have persistently higher-than-expected firearm mortality.
Weighted Mean SIR = 0.97	Average population-level excess risk	At the county population level, overall risk is near expected; high-risk tracts are small in population share.

Assessing Overlap Among Predictors

Variable	**G-VIF**	***df***	**GVIF^(1/(2*Df))**
ADI	1.06	4	1.01
Vegetation (NDVI)	1.50	1	1.23
Park Accessibility	1.26	1	1.12
Urban-Rural Labels (GHS-MOD)	1.57	7	1.03
Built Environment (GHS-BUILT)	1.49	1	1.22
Light (VIIRS)	1.48	1	1.22

Bayesian Spatiotemporal Modeling

Poisson BYM2 + RW1 Model Capturing Spatial and Temporal Dependence

Component	Equation	Description
Likelihood	$Y_{it} \sim Poisson(E_{it}\,\theta_{it})$	Counts in block group (i) and year (t) follow a Poisson with expected (E_{it}) and relative risk (θ_{it}).
Link	$\log(\theta_{it}) = \mathbf{x}_{it}^{\top}\boldsymbol{\beta} + u_i + v_i + \delta_t$	Log–relative risk is a function of covariates, spatial effects (u_i) structured, (v_i) unstructured, and temporal effect (δ_t).
Spatial (BYM2)	$u_i + v_i = \frac{1}{\sqrt{\tau}}\left(\sqrt{\phi}\, s_i + \sqrt{1-\phi}\, \epsilon_i\right)$	BYM2 mixes ICAR (s_i) and IID (ϵ_i) with mixing ($\phi \in [0,1]$); (τ) is overall precision.
Temporal (RW1)	$\delta_t \sim \mathcal{N}(\delta_{t-1},\, \tau_t^{-1})$	First-order random walk over years.
Priors	$\tau \sim (U = 1,\ \alpha = 0.01)$ $\phi \sim (0.5,\ 0.5)$ $\tau_t \sim (U = 1,\ \alpha = 0.01)$ $\boldsymbol{\beta} \sim \mathcal{N}(\mathbf{0},\ 10^6\mathbf{I})$	Penalized Complexity (PC) priors shrink toward the simpler base models (no spatial structure / no temporal wiggle); weakly-informative Normal prior on fixed effects.

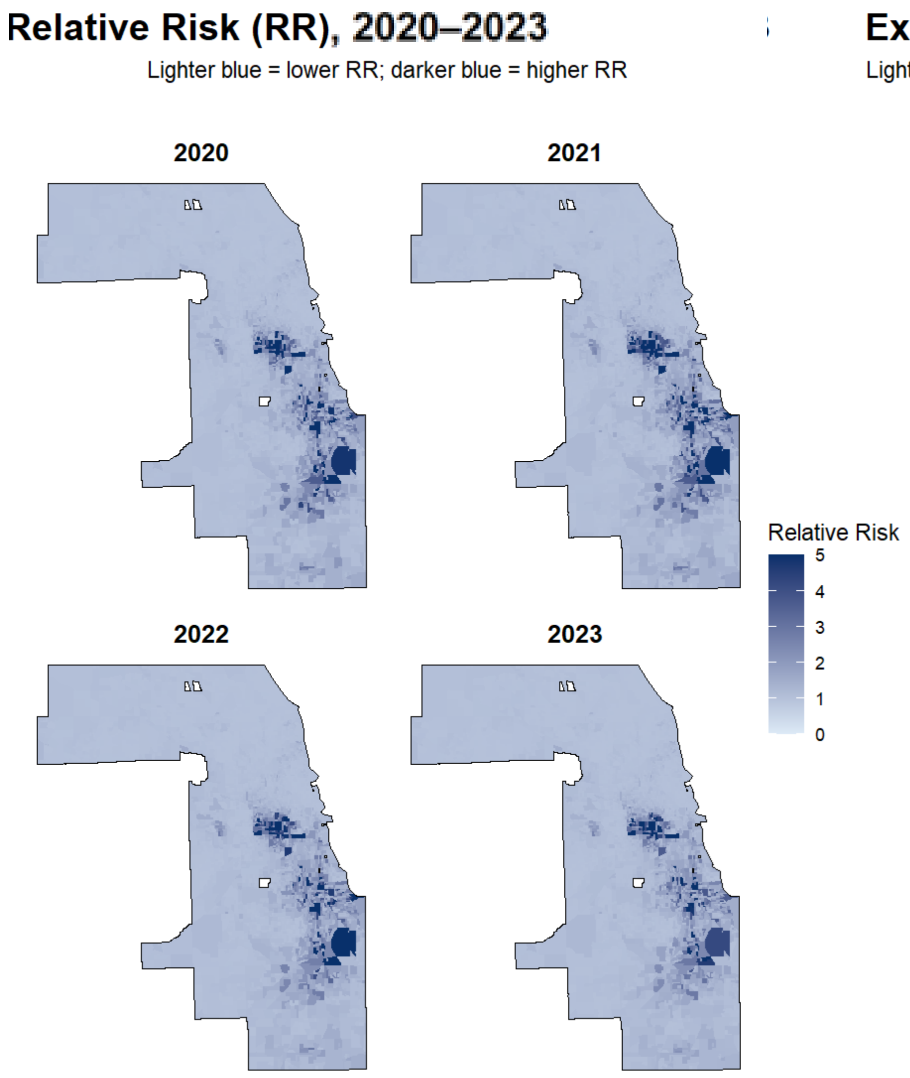
Model Specification and Fit

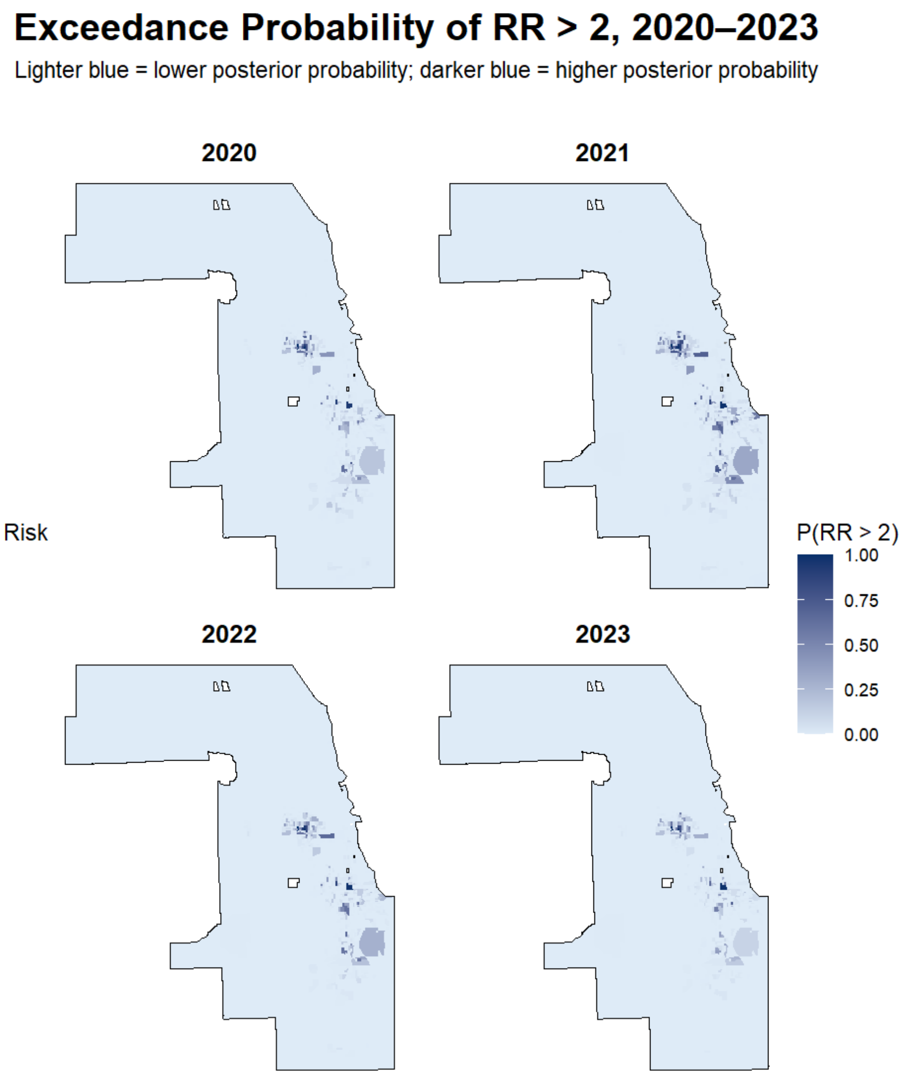
Model	Specification	Equation	Interpretation
M_0	IID space + RW1 time (null)	$\log(\theta_{it}) = \alpha + u_i + \delta_t$	Extremely poor fit
M_1	BYM2 + RW1	$\log(\theta_{it}) = \alpha + (u_i + v_i) + \delta_t$	Significant improvement (ΔDIC = –3,870)
M_2	+ Year + Urbanization	$\log(\theta_{it}) = \alpha + \beta_1 t + \sum_j \gamma_j U_{ij} + (u_i + v_i) + \delta_t$	Small significant improvement (ΔDIC = –40)
M_3	+ Natural environment (NDVI, acres)	$\log(\theta_{it}) = \alpha + \beta_1 t + \beta_2 \mathrm{NDVI}_i^* + \beta_3 A_i^* + (u_i + v_i) + \delta_t$	No improvement
M_4	+ Built + Light environment	$\log(\theta_{it}) = \alpha + \beta_1 t + \beta_2 \mathrm{NDVI}_i^* + \beta_3 A_i^* + \beta_4 B_i + \beta_5 L_i + (u_i + v_i) + \delta_t$	Small insignificant improvement (ΔDIC ≈ –17)
M_5	+ ADI quintile (Full)	$\log(\theta_{it}) = \alpha + \beta_1 t + \beta_2 \mathrm{NDVI}_i^* + \beta_3 A_i^* + \beta_4 B_i + \beta_5 L_i + \sum_k \eta_k \mathrm{ADI}_i^Q + (u_i + v_i) + \delta_t$	Best fit overall (lowest DIC / WAIC, highest MLIK)

Model Results

Variable	Mean	95% CrI
(Intercept)	−2.257	−3.482, −1.031
Year	0.004	−0.122, – 0.131
GHS-SMOD (Ref. Urban Core)		
Suburban	0.093	-0.232, 0.418
Transition (Semi-Dense Urban Cluster)	-0.633	-0.967, -0.256
Rural	-0.007	-0.483, 0.468
Natural Environment		
NDVI (z)	0.013	−0.073, – 0.108
Park Acreage within 15 min (z)	0.027	−0.036, – 0.088
Physical Environment		
Built Environment Index	−1.566	−2.622, −0.512
Nighttime Light Index	2.364	1.332, 3.404
Area Deprivation Index (Ref. Q1)		
Q2	0.596	0.360, 0.833
Q3	0.944	0.712, 1.183
Q4	1.053	0.810, 1.302
Q5	1.379	1.130, 1.637

Spatial Patterns of Relative Risk



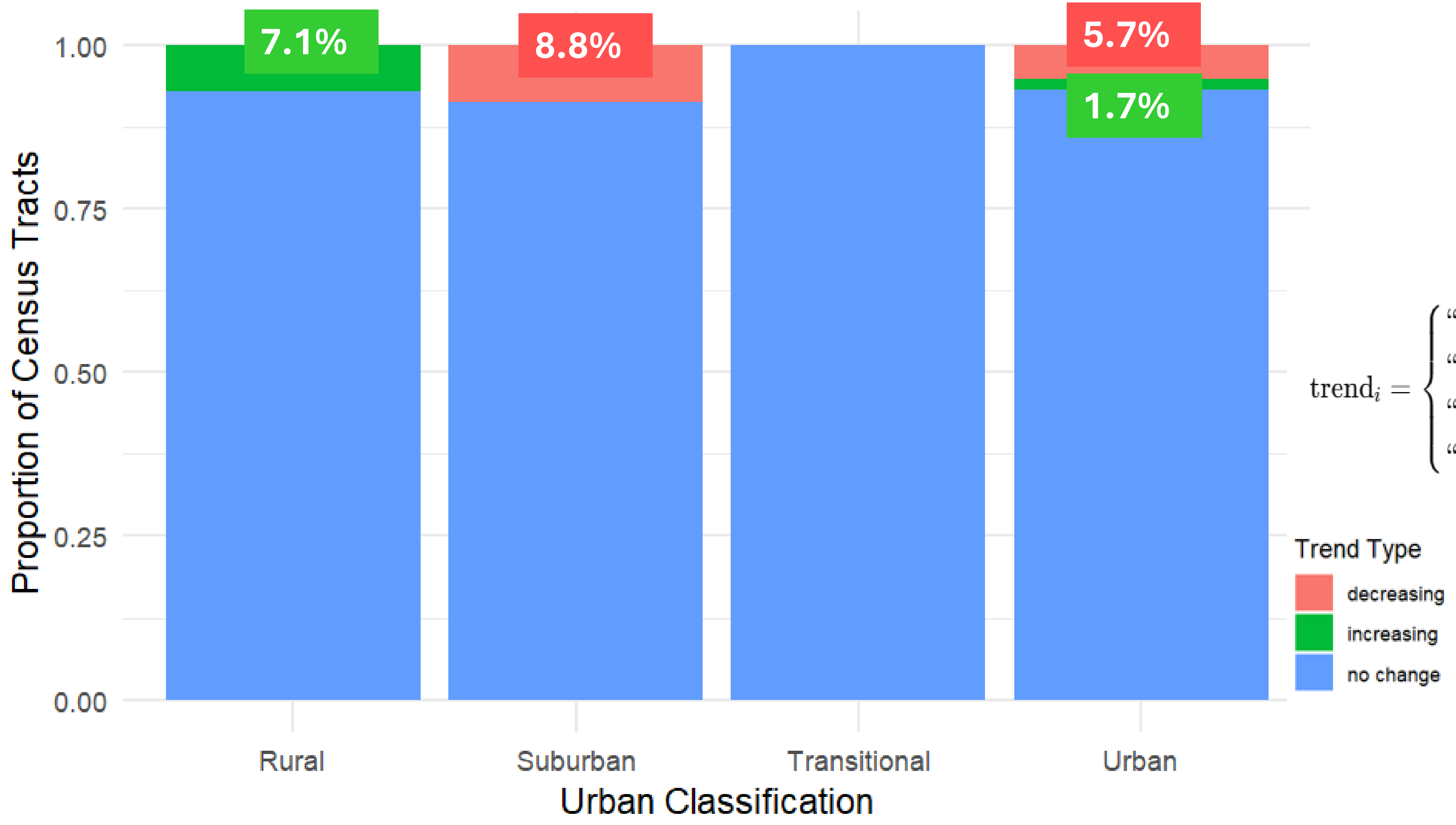


$$RR_{it} = \alpha_i + \beta_{1i} \cdot t + \beta_{2i} \cdot t^2 + \epsilon_{it}$$

$$P(RR_{it} > 2) = 1 - \Phi\left(\frac{\log(2) - \mu_{it}}{\sigma_{it}}\right)$$

Derived from INLA-BYM2-RW1 Spatiotemporal Model (M5)

Temporal Trends and Urban Gradients



$$\text{trend}_i = \begin{cases} \text{``sharply increasing''}, & \beta_{1i} > 0, \beta_{2i} > 0, p < 0.05 \\ \text{``increasing''}, & \beta_{1i} > 0, p < 0.05 \\ \text{``decreasing''}, & \beta_{1i} < 0, p < 0.05 \\ \text{``no change''}, & \text{otherwise} \end{cases}$$

Conclusions and Future Directions

- This study integrates environmental, socioeconomic, and urban form indicators within a Bayesian spatiotemporal framework to model firearm injury risk at a fine geographic scale
- Findings reveal persistent high-risk clusters concentrated in socially disadvantaged urban areas
- Environmental indicators (built density, light intensity) increase the explanatory power of spatial firearm mortality risk, demonstrating that environmental exposures are important to understanding beyond social deprivation alone
  - Built environment (structure): physical infrastructure and connectedness ***buffer*** risk
  - Light environment (activity): Where social and economic energy ***exposes*** risk
- Future work will disaggregate the manner of death
- Plan to integrate additional environmental exposure measures (e.g., heat, air quality, accessibility) and perhaps death narratives

Policy & Prevention Implications

- Highlights the importance of place-based violence prevention that integrates both social and environmental interventions
- High deprivation and high light/built density areas identify target zones for environmental remediation and community investment
- Supports shifting prevention from individual-level to structural and environmental determinants of firearm injury
- Spatially explicit modeling ***must*** guide equitable resource allocation, particularly in historically underinvested neighborhoods
    - Vegetation and park proximity alone are not sufficient to reduce risk in contexts of socio-structural disadvantage

INVESTIGATING SPATIAL STRUCTURES in Urban Environments FOR LEGAL AND POLICY INSIGHTS

LAB

Thank you for coming!

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